类目

Mathematics / Dynamical systems

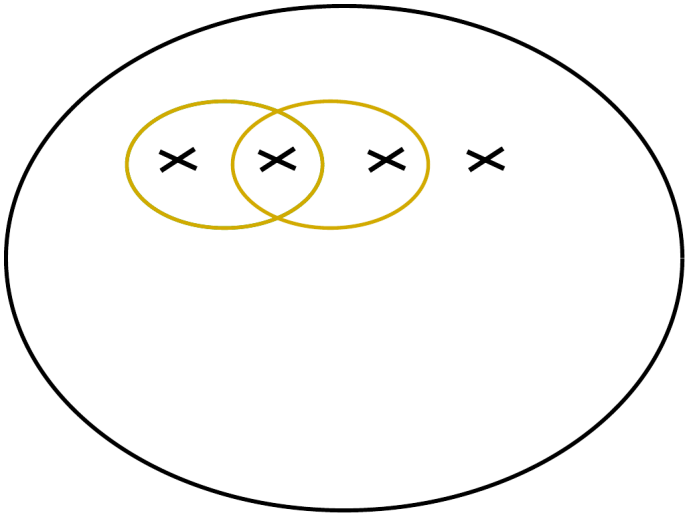
题目

An oriented sphere S with four marked points is presented by the planar chart bounded by the outer black oval. The disk outside that oval completes the sphere and contains no additional marked point. The four black crosses are the marked points p_1, p_2, p_3, p_4, indexed from left to right, and all isotopies fix them pointwise. Each cross is a distinguished puncture, and the surface has no boundary. The two yellow traces are embedded essential simple closed curves in $S \setminus \{p_1, p_2, p_3, p_4\}$. Every meeting of their traces is a transverse intersection; no over-under information or tangency is intended. Each curve follows its complete yellow trace, and the line thickness has no mathematical weight.

Let a be the yellow curve whose bounded planar side contains p_1, and let b be the other yellow curve. Curves are considered up to isotopy relative to the four marked points. Orient the planar chart counterclockwise, thereby fixing the orientation of S. For an essential simple closed curve c, let T_c denote the right Dehn twist about c, supported in a small annulus disjoint from all marks. Mapping classes act on the left, and composition is read right-to-left. Define

$$f = T_a \circ T_b^{-1} \in \mathrm{PMod}(S, \{p_1, p_2, p_3, p_4\}).$$

For a pseudo-Anosov mapping class h, write $\lambda(h) > 1$ for the stretch factor that multiplies its unstable measured foliation. Determine the exact value of $\lambda(f)$. Report one real algebraic number in simplest radical form. Include no matrix, polynomial, intersection number, mapping-class label, or categorical verdict as an additional answer component.



答案

$$\lambda(f) = 3 + 2\sqrt{2}$$

解析

The bounded sides of the two yellow curves contain the marked-point sets

$$\operatorname{int}(a) \cap \{p_1, p_2, p_3, p_4\} = \{p_1, p_2\}, \qquad \operatorname{int}(b) \cap \{p_1, p_2, p_3, p_4\} = \{p_2, p_3\}.$$

Following both complete traces shows that they meet transversely at the two boundary points of their overlap:

$$\mu := i(a, b) = |a \cap b| = 2.$$

Their union separates the punctures into the left-only, overlap, right-only, and exterior regions, so the complement has four components:

$$S \setminus (a \cup b) = D_1^* \sqcup D_2^* \sqcup D_3^* \sqcup D_4^*.$$

Each D_r^* is a disk containing exactly the marked point p_r. Hence the two one-curve multicurves fill the punctured sphere:

$$\{a\} \cup \{b\} \text{ fills } S \setminus \{p_1, p_2, p_3, p_4\}.$$

The opposite twist signs and the filling property satisfy the hypotheses of Penner's construction, which makes the prescribed product pseudo-Anosov:

$$f = T_a T_b^{-1} \quad \text{is pseudo-Anosov.}$$

Smoothing the two intersections in the Penner direction gives a common invariant bigon track. On its two weight coordinates, the negative twist about b acts by

$$B\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u \\ \mu u + v \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ \mu & 1 \end{pmatrix}.$$

The subsequent positive twist about a adds the b-weight across every intersection:

$$A\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u + \mu v \\ v \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix}.$$

Because composition is right-to-left, the transition matrix for f on this invariant cone is

$$M = AB = \begin{pmatrix} 1 + \mu^2 & \mu \\ \mu & 1 \end{pmatrix}.$$

Substituting the geometric intersection number produces a positive integral matrix:

$$M = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \qquad \det M = 1, \qquad \operatorname{tr} M = 6.$$

Its characteristic polynomial is therefore

$$\chi_M(t) = t^2 - 6t + 1.$$

The two eigenvalues are reciprocal, and the Perron root is the one exceeding one:

$$\operatorname{Spec}(M) = \{3 + 2\sqrt{2},\ 3 - 2\sqrt{2}\}, \qquad \rho(M) = 3 + 2\sqrt{2}.$$

Penner's invariant-track theorem identifies this Perron root with the stretch factor, giving the requested exact output:

$$\lambda(f) = 3 + 2\sqrt{2}.$$

类目

Mathematics / Hypergraph theory

题目

A 3-uniform hypergraph H has the ordered vertex set $(1,2,3,3',4,4')$. Its two-part representation gives a complete 3-graph on $1,2,3,4$ followed by a crossed-blowup representation. A gray triangular face represents its three incident vertices as a hyperedge. A colored link convention is also used: if a colored vertex is incident with a segment of the same color joining two other vertices, those three vertices form a hyperedge. The color of a segment therefore identifies its link vertex, and a black segment carries no extra colored-link hyperedge. The gray skeletal faces in the right member represent every hyperedge containing both vertices 1 and 2 together with one of $3,3',4,4'$. The red link class has link vertex 1 and joins the two indicated red parts. The yellow link class has link vertex 2 and joins the two indicated yellow parts. No multiplicities occur. For $x_i \geq 0$ with $x_1+x_2+x_3+x_{3'}+x_4+x_{4'}=1$, define $p_H(x)=\sum_{\{i,j,k\}\in E(H)} x_i x_j x_k$ and $\lambda(H)=\max p_H(x)$. Use the coordinate order $(x_1,x_2,x_3,x_{3'},x_4,x_{4'})$. Determine the ordered pair consisting of $\lambda(H)$ and the complete set of maximizing weights, written as one affine family with parameter $0\leq t\leq 1$. The scalar is the first component. The second component is the set of six-coordinate vectors in the stipulated coordinate order. The priming convention is the one shown in the crossed-blowup member. Hyperedges are unordered three-element sets, so a segment contributes exactly through its stated colored link vertex. The gray and colored conventions apply uniformly to every indicated incidence. They are distinct. The two link colors identify separate incidence rules throughout the representation.

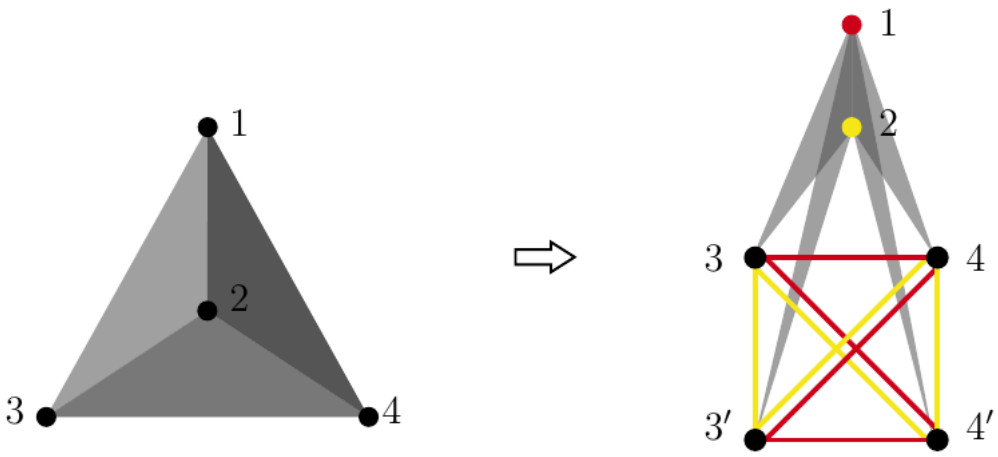


Figure 1: K_4^3 and $K_4^3 \boxplus \{3,4\}$.

答案

$$\left(\frac{1}{16},\left\{\left(\frac{1}{4},\frac{1}{4},\frac{t}{4},\frac{1-t}{4},\frac{1-t}{4},\frac{t}{4}\right):0\leq t\leq 1\right\}\right)$$

解析

Set $r = x_1$, $s = x_2$, $a = x_3$, $b = x_4$, $c = x_{3'}$, and $d = x_{4'}$. The gray faces contribute the terms containing both r and s, and the two colored link classes contribute the two displayed complete bipartite products.

$$p_H = rs(a+b+c+d) + r(a+c)(b+d) + s(a+b)(c+d)$$

Put $S = a+b+c+d$ and $q = r+s$, so the simplex condition gives the complementary relation.

$$q + S = 1$$

The red bipartite product has two nonnegative factors with sum S.

$$(a+c)(b+d) \le \frac{S^2}{4}$$

The yellow bipartite product has the same total factor sum.

$$(a+b)(c+d) \le \frac{S^2}{4}$$

The first product is bounded by balancing the two gray-link coordinates.

$$rs \le \frac{q^2}{4}$$

Substitution of these three bounds reduces the cubic to a function of the two complementary totals.

$$p_H \le \frac{q^2 S}{4} + \frac{qS^2}{4}$$

The simplex relation combines the two terms.

$$p_H \le \frac{qS(q+S)}{4}$$

$$p_H \le \frac{qS}{4}$$

The final two-factor product is maximal only when the two totals agree.

$$qS \le \frac{1}{4}$$

Consequently the Lagrangian has the upper bound.

$$p_H \le \frac{1}{16}$$

Equality requires equality in every preceding balancing inequality.

$$r = s = \frac{1}{4} \quad \text{and} \quad S = \frac{1}{2}$$

The red equality condition fixes its two displayed parts to equal total weight.

$$a + c = b + d = \frac{1}{4}$$

The yellow equality condition gives the other two displayed parts.

$$a+b=c+d=\frac{1}{4}$$

Solving these two link equalities yields the crossed pairing normal form.

$$a=d \quad \text{and} \quad b=c$$

Writing $a=d=t/4$ gives the complete normalized equality family.

$$x(t)=\left(\frac{1}{4},\frac{1}{4},\frac{t}{4},\frac{1-t}{4},\frac{1-t}{4},\frac{t}{4}\right)$$

Nonnegativity is exactly the interval condition on the parameter.

$$0\le t\le 1$$

Each vector in this family makes all balancing inequalities equalities, so the requested ordered pair is $\left(\frac{1}{16},\{x(t):0\le t\le 1\}\right)$.

类目

> **Physics / Quantum information**

题目

Two commuting-stabilizer Wen plaquette layers have Abelian charge group $\mathcal{A} = \mathbb{F}_2^4$ in the ordered basis (e_1, m_1, e_2, m_2); fusion is vector addition. When a charge vector enters a braiding phase, use its 0-1 integer representative and

$$K = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{pmatrix}, \qquad b(a,t) = \exp\!\left(2\pi i\, a^{T} K^{-1} t\right).$$

The symmetry σ exchanges $e_j \leftrightarrow m_j$ in each layer, while ρ exchanges the two layer labels and preserves electric or magnetic type. If incident branch-cut matrices are L_1 and L_2, their outgoing matrix is $L = L_1 L_2$, with the order fixed by the incoming lines in the diagram.

For any twofold symmetry matrix M, charges on its defect are identified modulo $\mathrm{Im}(I + M)$. For the Figure-selected incoming pair define

$$\mathcal{N} = \mathrm{Im}(I + L_1) + \mathrm{Im}(I + L_2), \qquad \mathcal{D} = \mathrm{Im}(I + L).$$

The bare incoming defects have zero attached-charge classes. Their neutral outgoing channels are $\mathcal{N}/\mathcal{D}$; a junction charge $s \in \mathcal{A}$ translates every outgoing representative by s. Construct the unique rank-two reduced-row-echelon matrix Q with $\ker Q = \mathcal{D}$ and pivot columns $1, 2$. For $u = Qt = (u_1, u_2)^{T}$, fix the outgoing species convention by $[t] \mapsto \xi_{u_1 + 2u_2}$.

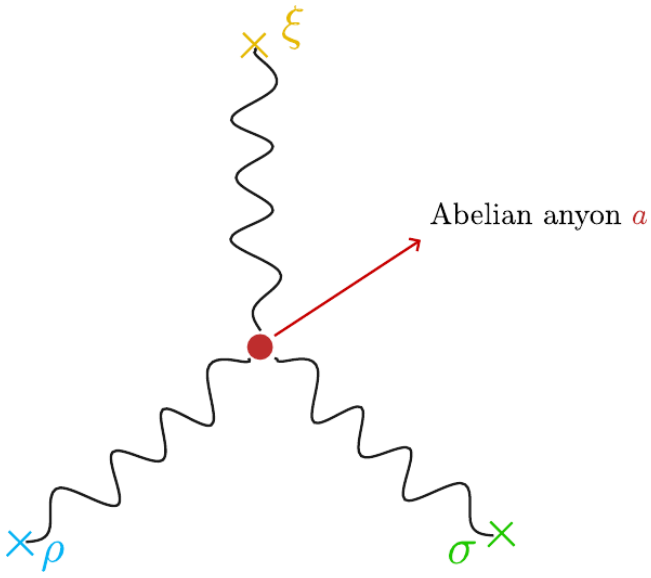
Use the two invariant probe charges

$$q_1 = m_1 + e_2 = (0,1,1,0)^{T}, \qquad q_2 = e_1 + m_2 = (1,0,0,1)^{T}.$$

The relative eigenvalue of probe loop j caused by s is $\chi_j(s) = b(q_j, s)$. On the two local junction qubits, the loop operators have the unsimplified words

$$\widehat{\Omega}_1 = Z_1 Z_2 \mathcal{S}_1, \qquad \widehat{\Omega}_2 = X_1 X_2 \mathcal{S}_2,$$

where X, Y, Z are the standard Pauli matrices, each $\mathcal{S}_j$ commutes with both local Pauli products, $\mathcal{S}_1$ and $\mathcal{S}_2$ commute, and $\mathcal{S}_1 \mathcal{S}_2$ has eigenvalue $+1$ in the stabilizer ground sector, and $J = Y_1 Y_2$ has eigenvalue $+1$ for the visibly occupied red junction and -1 for an unoccupied junction. The Figure is to be used only to read the incident labels, their connectivity, and which of these two junction branches is present. Determine the exact fusion product of the two incoming bare defects.



答案

$$\sigma_0 \times \rho_0 = \xi_1 + \xi_2$$

解析

The visible incoming labels select the within-layer exchange and the layer exchange. In the stated ordered basis their matrices are

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Multiplying the incident actions gives the automorphism carried by the visible outgoing line:

$$\Lambda = SR = RS = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \Lambda^2 = I.$$

The first image subgroup follows from a finite-field reduction:

$$I + S = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad \mathcal{A}_\sigma = \langle (1,1,0,0)^T, (0,0,1,1)^T \rangle, \qquad \operatorname{rank} \mathcal{A}_\sigma = 2.$$

The second incoming line has a distinct image subgroup:

$$I+R=\begin{pmatrix}1&0&1&0\\0&1&0&1\\1&0&1&0\\0&1&0&1\end{pmatrix},\qquad \mathcal{A}_\rho=\langle(1,0,1,0)^T,(0,1,0,1)^T\rangle,\qquad \operatorname{rank}\mathcal{A}_\rho=2.$$

Concatenating independent generators constructs the subgroup absorbable on the two incoming lines:

$$G_{\mathcal{N}}=\begin{pmatrix}1&0&1&0\\1&0&0&1\\0&1&1&0\\0&1&0&1\end{pmatrix},\qquad \mathcal{N}=\operatorname{colspan}_{\mathbb{F}_2}G_{\mathcal{N}}.$$

Its reduction has three pivots, so the incoming sum subgroup contains eight charges:

$$\operatorname{rref}_{\mathbb{F}_2}(G_{\mathcal{N}}^T)=\begin{pmatrix}1&0&0&1\\0&1&0&1\\0&0&1&1\\0&0&0&0\end{pmatrix},\qquad \operatorname{rank}\mathcal{N}=3,\qquad |\mathcal{N}|=8.$$

The outgoing equivalence is obtained from the composed action:

$$I+\Lambda=\begin{pmatrix}1&0&0&1\\0&1&1&0\\0&1&1&0\\1&0&0&1\end{pmatrix}.$$

The column space of this outgoing operator is

$$\mathcal{D}=\langle d_1,d_2\rangle,\qquad d_1=(1,0,0,1)^T,\qquad d_2=(0,1,1,0)^T,\qquad \operatorname{rank}\mathcal{D}=2.$$

Both outgoing generators belong to the incoming sum, as the explicit decompositions show:

$$d_1=(1,1,0,0)^T+(0,1,0,1)^T,\qquad d_2=(1,1,0,0)^T+(1,0,1,0)^T,\qquad \mathcal{D}\subset\mathcal{N}.$$

Solving the stated kernel and pivot conditions gives the quotient-coordinate map

$$Q=\begin{pmatrix}1&0&0&1\\0&1&1&0\end{pmatrix},\qquad Qd_1=Qd_2=0,\qquad \ker Q=\mathcal{D}.$$

Applying it to the incoming sum produces the neutral channel set in quotient coordinates:

$$H=Q(\mathcal{N})=\{(0,0)^T,(1,1)^T\},\qquad |H|=2^{\operatorname{rank}\mathcal{N}-\operatorname{rank}\mathcal{D}}=2.$$

The braiding data now connect a junction charge to the two quotient coordinates. Direct inversion of K gives

$$2\begin{pmatrix}q_1^TK^{-1}\\q_2^TK^{-1}\end{pmatrix}=\begin{pmatrix}1&0&0&1\\0&1&1&0\end{pmatrix}=Q.$$

Writing $u(s)=Qs$ therefore turns the two relative loop characters into

$$\chi_1(s)=(-1)^{u_1(s)},\qquad \chi_2(s)=(-1)^{u_2(s)}.$$

The microscopic value of their product must still be derived from the two raw loop words:

$$\begin{aligned}\widehat{\Omega}_1\widehat{\Omega}_2 &= Z_1Z_2X_1X_2\mathcal{S}_1\mathcal{S}_2\\ &= (Z_1X_1)(Z_2X_2)\mathcal{S}_1\mathcal{S}_2.\end{aligned}$$

Standard Pauli multiplication supplies one phase at each local qubit:

$$(Z_1X_1)(Z_2X_2) = (iY_1)(iY_2) = -Y_1Y_2 = -J.$$

The stabilizer eigenvalue and the occupied red junction read from the Figure then fix the relative product:

$$\begin{aligned}\chi_1(s)\chi_2(s) &= \mathrm{eig}(-J\mathcal{S}_1\mathcal{S}_2)\\ &= -(+1)(+1) = -1.\end{aligned}$$

Combining this microscopic sign with the character formula yields a nontrivial affine constraint:

$$(-1)^{u_1(s)+u_2(s)} = -1, \qquad u_1(s)+u_2(s) = 1 \pmod 2.$$

Thus the projective translation is odd, but the loop branch does not choose one charge class:

$$u(s) = Qs \in \{(1,0)^T, (0,1)^T\}.$$

For either allowed value, translating the neutral quotient image gives the same two actual channel coordinates:

$$u(s) + H = \{(1,0)^T, (0,1)^T\}.$$

The prescribed index rule finally identifies both channels:

$$(1,0)^T \mapsto \xi_{1+2\cdot 0} = \xi_1, \qquad (0,1)^T \mapsto \xi_{0+2\cdot 1} = \xi_2.$$

The two Figure-selected bare defects consequently have the exact fusion product

$$\sigma_0 \times \rho_0 = \xi_1 + \xi_2.$$

类目

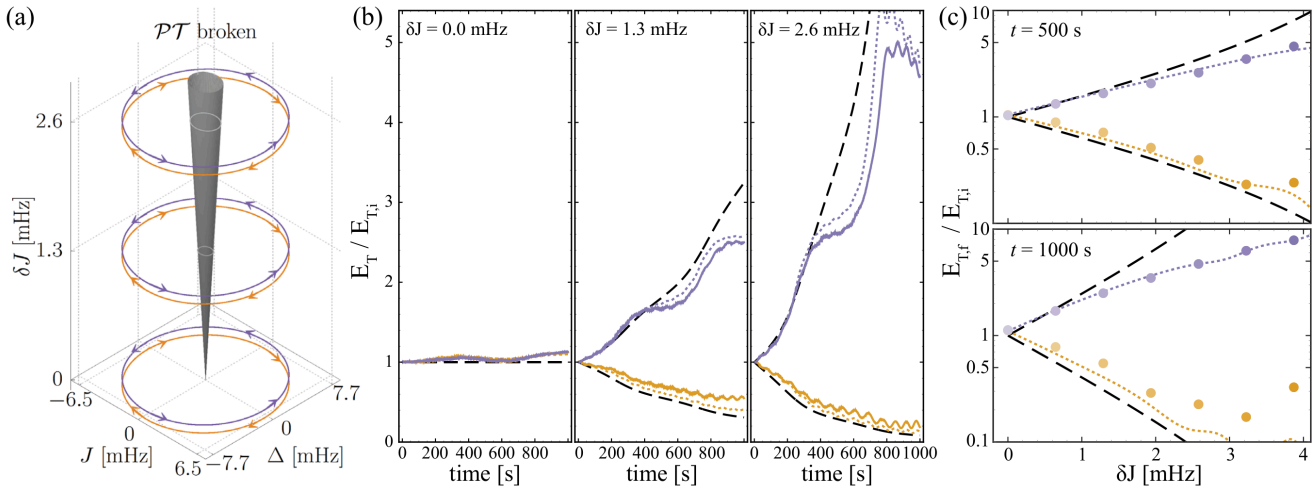
Physics / Non-Hermitian geometric phases

题目

A pair of feedback-coupled oscillators realizes the two-site non-Hermitian Hamiltonian $H/h = \begin{pmatrix} -\Delta & J+\delta J \\ J-\delta J & \Delta \end{pmatrix}$, with real parameters and eigenvalues $\varepsilon_s = s\sqrt{\Delta^2 + J^2 - \delta J^2}$ for $s \in \{+1, -1\}$. Work in the nondegenerate adiabatic domain $\Delta^2 + J^2 > \delta J^2$, fix a positive constant δJ, and use the complete oriented loop shown in panel (a) as the path in the J-Δ plane. The loop is the ellipse $\Delta = \Delta_0 \cos\theta$, $J = J_0 \sin\theta$ with $0 \le \theta \le 2\pi$; the image supplies its orientation and shows that the exceptional cone lies inside it. The source convention gives the Berry-curvature field $\mathbf{\Omega}_s = si(\Delta, J, \delta J)/[2(\Delta^2 + J^2 - \delta J^2)^{3/2}]$. Treat the path as fully adiabatic, keep the band sign s symbolic, and define the complete elliptic integral by $\mathrm{Pi}(n \mid m) = \int_0^{\pi/2} d\theta/[(1 - n\sin^2\theta)\sqrt{1 - m\sin^2\theta}]$. Using only the displayed geometry, these model definitions, Stokes' theorem on a continuous surface, and exact symbolic manipulation, determine the single closed-form non-Hermitian Berry phase ϕ_s acquired on this oriented loop.

Traverse the ellipse once with increasing θ and use the continuous biorthogonal eigenvector branch specified in the stem. The Berry phase is defined modulo 2π.

State one dimensionless expression, with all frequency parameters in a common unit.



答案

$$\phi_s = s\pi - si\,\frac{2\delta J J_0}{\Delta_0\sqrt{\Delta_0^2 - \delta J^2}}\,\mathrm{Pi}\left(1 - \frac{J_0^2}{\Delta_0^2}\,\middle|\,\frac{\Delta_0^2 - J_0^2}{\Delta_0^2 - \delta J^2}\right) \pmod{2\pi}$$

解析

The oriented loop determines which side of the cylindrical deformation is traversed, while the cone enclosure guarantees that this deformation avoids the singular exceptional surface. Let the band sign be s.

$$\mathbf{\Omega}_s(\boldsymbol{\lambda}) = \frac{si(\Delta, J, \delta J)}{2(\Delta^2 + J^2 - \delta J^2)^{3/2}}$$

The curvature is continuous on the allowed cylinder, so Stokes' theorem converts the line phase into a surface flux.

$$\boldsymbol{\lambda}(\theta, \delta J') = (\Delta_0 \cos\theta, J_0 \sin\theta, \delta J')$$

The side joins the physical loop to the regular base, and its two tangent vectors give the oriented element.

$$\begin{aligned} \partial_\theta \boldsymbol{\lambda} &= (-\Delta_0 \sin\theta, J_0 \cos\theta, 0), \\ \partial_{\delta J'} \boldsymbol{\lambda} &= (0, 0, 1), \\ \partial_\theta \boldsymbol{\lambda} \times \partial_{\delta J'} \boldsymbol{\lambda} &= (J_0 \cos\theta, \Delta_0 \sin\theta, 0). \end{aligned}$$

Contracting the curvature with this oriented element gives

$$\mathbf{\Omega}_s \cdot (\partial_\theta \boldsymbol{\lambda} \times \partial_{\delta J'} \boldsymbol{\lambda}) = \frac{si \Delta_0 J_0}{2[\Delta_0^2 \cos^2\theta + J_0^2 \sin^2\theta - \delta J'^2]^{3/2}}$$

Only the first two curvature components contribute to the side flux, producing a scalar integrand with the band sign intact.

$$\phi_s = \int_0^{2\pi} d\theta \int_{\delta J}^{0} d\delta J' \, \frac{si \Delta_0 J_0}{2[\Delta_0^2 \cos^2\theta + J_0^2 \sin^2\theta - \delta J'^2]^{3/2}}$$

The upper-to-lower height limits encode the oriented deformation; reversing the loop would reverse these limits.

$$\int_{\delta J}^{0} \frac{d\delta J'}{(A - \delta J'^2)^{3/2}} = -\frac{\delta J}{A\sqrt{A - \delta J^2}}, \qquad A = \Delta_0^2 \cos^2\theta + J_0^2 \sin^2\theta$$

The elementary antiderivative removes the auxiliary coordinate and contributes the orientation minus sign.

$$\phi_s = -si \, \frac{\Delta_0 J_0 \delta J}{2} \int_0^{2\pi} \frac{d\theta}{A\sqrt{A - \delta J^2}}$$

The remaining integral is periodic under quadrant reflections because it depends only on squared trigonometric functions.

$$\phi_s = -si \, 2\Delta_0 J_0 \delta J \int_0^{\pi/2} \frac{d\theta}{A\sqrt{A - \delta J^2}}$$

Four identical quadrants replace the full interval by four times the first quadrant, yielding the factor shown.

$$A = \Delta_0^2 \left[1 - \left(1 - \frac{J_0^2}{\Delta_0^2}\right) \sin^2\theta\right]$$

Factoring the detuning scale exposes the rational elliptic denominator.

$$A - \delta J^2 = (\Delta_0^2 - \delta J^2)\left[1 - \frac{\Delta_0^2 - J_0^2}{\Delta_0^2 - \delta J^2}\sin^2\theta\right]$$

Factoring the residual scale exposes the square-root elliptic parameter.

$$n = 1 - \frac{J_0^2}{\Delta_0^2}, \qquad m = \frac{\Delta_0^2 - J_0^2}{\Delta_0^2 - \delta J^2}$$

These two dimensionless combinations match the characteristic and parameter in the normalized integral.

$$\frac{1}{A\sqrt{A - \delta J^2}} = \frac{1}{\Delta_0^2\sqrt{\Delta_0^2 - \delta J^2}}\,\frac{1}{(1 - n\sin^2\theta)\sqrt{1 - m\sin^2\theta}}$$

Substitution separates a dimensionless kernel from the frequency prefactor.

$$\int_0^{\pi/2} \frac{d\theta}{(1 - n\sin^2\theta)\sqrt{1 - m\sin^2\theta}} = \mathrm{Pi}(n \mid m)$$

The source normalization identifies the kernel with the complete elliptic integral of the third kind.

$$\phi_s = -si\,\frac{2\Delta_0 J_0 \delta J}{\Delta_0^2\sqrt{\Delta_0^2 - \delta J^2}}\mathrm{Pi}(n \mid m)$$

Cancelling one detuning factor gives the compact prefactor while preserving dimensions.

$$\Delta\phi_s = -si\,\frac{2\delta J J_0}{\Delta_0\sqrt{\Delta_0^2 - \delta J^2}}\mathrm{Pi}\left(1 - \frac{J_0^2}{\Delta_0^2}\,\middle|\,\frac{\Delta_0^2 - J_0^2}{\Delta_0^2 - \delta J^2}\right)$$

This side flux is the purely imaginary correction. It vanishes with the non-reciprocal coupling, while the complete phase retains the real spinor baseline derived below.

The continuously transported two-component eigenvector also winds once, contributing the baseline

$$\phi_s^{(0)} = s\pi \pmod{2\pi}.$$

Adding the winding baseline to the side flux gives

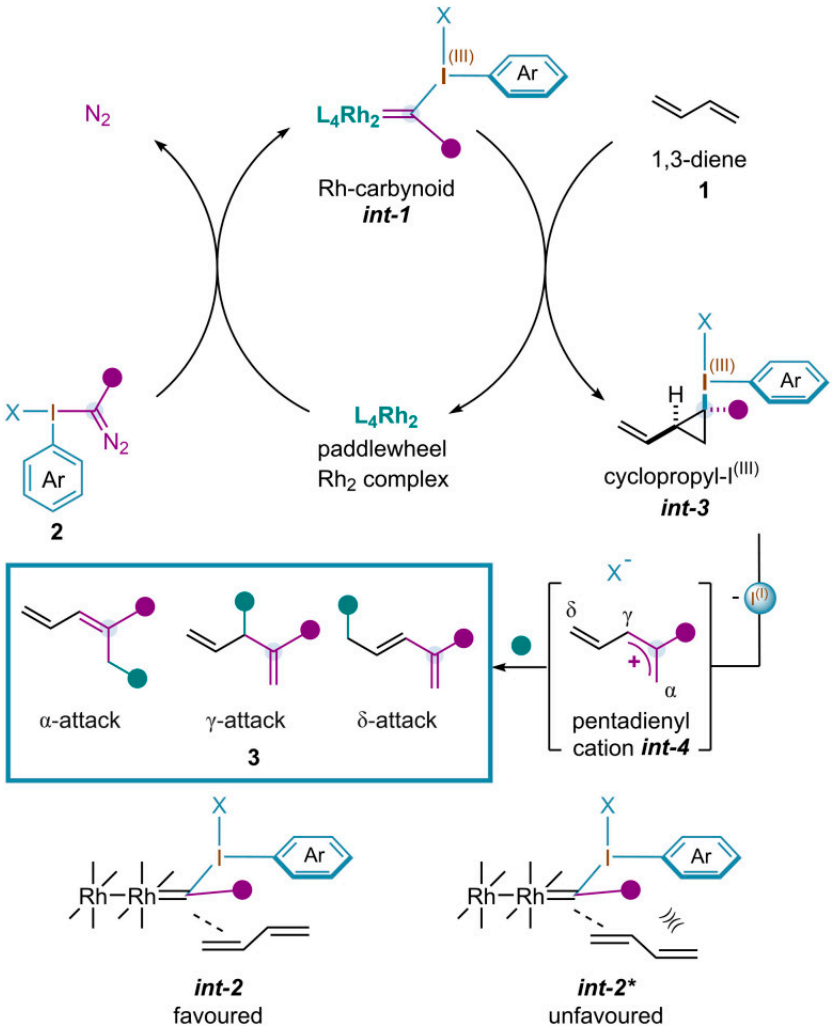
$$\phi_s = s\pi - si\,\frac{2\delta J J_0}{\Delta_0\sqrt{\Delta_0^2 - \delta J^2}}\mathrm{Pi}\left(1 - \frac{J_0^2}{\Delta_0^2}\,\middle|\,\frac{\Delta_0^2 - J_0^2}{\Delta_0^2 - \delta J^2}\right) \pmod{2\pi}.$$

类目

Chemistry / Skeletal editing and reaction mechanisms

题目

A dirhodium-catalyzed carbynoid skeletal-editing reaction expands a conjugated four-carbon framework through single-carbon insertion and subsequent nucleophilic capture. Carbon-skeleton editing has become a valuable strategy for changing molecular architecture while preserving much of a complex substrate. Such transformations can shorten synthetic sequences, diversify advanced intermediates, and provide retrosynthetic disconnections that are unavailable through peripheral functionalization. Conjugated dienes are widespread building blocks in polymer chemistry, medicinal chemistry, natural-product synthesis, and cycloaddition chemistry. Catalytic control over their connectivity therefore has broad significance for selective synthesis and molecular design. The carbons of the displayed 1,3-diene carry left-to-right labels $\mathbf{C_1}$ through $\mathbf{C_4}$, and the carbynoid carbon bearing $\mathbf{R}$ is $\mathbf{C^*}$. Purple marks $\mathbf{R}$ attached to $\mathbf{C^*}$, while teal marks $\mathbf{Nu}$ at the carbon attacked by the nucleophile. Report the finite set of three neutral atom-mapped constitutional products obtained from the three electrophilic capture positions, retaining the carbon numbering and every carbon-carbon double bond.



答案

$$\left\{\begin{array}{c}\mathrm{CH_2^{(1)}=CH^{(2)}-CH^{(3)}=C^*(R)-CH_2^{(4)}Nu},\\ \mathrm{CH_2^{(1)}=CH^{(2)}-CH^{(3)}(Nu)-C^*(R)=CH_2^{(4)}},\\ \mathrm{NuCH_2^{(1)}-CH^{(2)}=CH^{(3)}-C^*(R)=CH_2^{(4)}}\end{array}\right\}$$

解析

Cyclopropanation uses the right-hand alkene of the numbered diene, so the starting carbon order is

$$\mathrm{C_1=C_2-C_3=C_4.}$$

The carbynoid carbon becomes the third vertex of the three-membered intermediate while retaining R. Its vertex map is

$$\mathrm{C_3-C^*(R)-C_4,}\qquad \mathrm{C_3-C_4\ closed.}$$

Disrotatory cleavage of the original $\mathrm{C_3-C_4}$ ring edge inserts $\mathrm{C^*}$ into the carbon chain. The opened order is

$$\mathrm{C_1-C_2-C_3-C^*(R)-C_4.}$$

Capture at the alpha terminus places Nu on $\mathrm{C_4}$ and forms the internal conjugated pair shown for that branch:

$$\mathrm{CH_2^{(1)}=CH^{(2)}-CH^{(3)}=C^*(R)-CH_2^{(4)}Nu.}$$

Capture at the gamma position places Nu on $\mathrm{C_3}$ and shifts the second double bond to $\mathrm{C^*=C_4}$:

$$\mathrm{CH_2^{(1)}=CH^{(2)}-CH^{(3)}(Nu)-C^*(R)=CH_2^{(4)}.}$$

Capture at the delta terminus places Nu on $\mathrm{C_1}$ and leaves the two displayed double bonds on $\mathrm{C_2=C_3}$ and $\mathrm{C^*=C_4}$:

$$\mathrm{NuCH_2^{(1)}-CH^{(2)}=CH^{(3)}-C^*(R)=CH_2^{(4)}.}$$

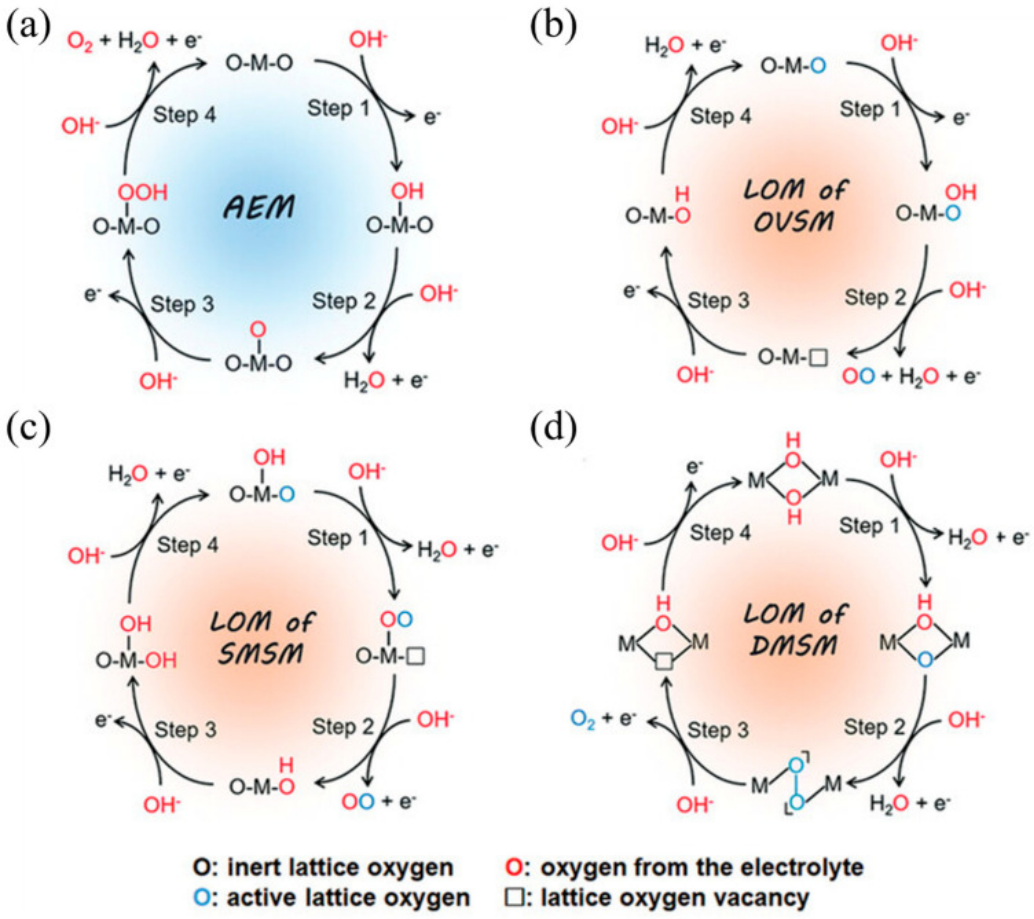
Each branch preserves all five carbon identities and closes carbon valence after one nucleophilic capture. The three atom-mapped graphs therefore constitute the requested finite set.

类目

Chemistry / Electrocatalysis and isotope tracing

题目

Alkaline water electrolysis converts renewable electricity into hydrogen while molecular oxygen forms at the anode. The oxygen evolution reaction controls a large fraction of the kinetic voltage loss in many alkaline electrolyzers. Mechanistically informed catalyst design can lower energy consumption, extend electrode lifetime, and support operation at industrial current density. Isotope tracing is widely applied in surface chemistry and heterogeneous catalysis because isotopes preserve elemental valence while marking distinct atomic reservoirs. These experiments can connect oxygen exchange, catalyst reconstruction, and catalytic turnover under operating conditions. In the cycle color convention, blue oxygen is active lattice oxygen, red oxygen comes from electrolyte hydroxide, and black oxygen is inert lattice oxygen. For the isotope experiment, every blue oxygen initially is $^{16}\mathrm{O}$, every red oxygen supplied as $^{18}\mathrm{OH}^{-}$ is $^{18}\mathrm{O}$, and water contains only $^{18}\mathrm{O}$. Let $\mathcal{C}[x, y]$ denote the catalyst with isotopes x and y at the two adjacent tracked lattice sites; all other catalyst atoms are spectators. One complete turnover follows panel (d), starts from $\mathcal{C}[^{16}\mathrm{O}, ^{16}\mathrm{O}]$, and refills each generated vacancy only from electrolyte hydroxide. Report the balanced isotope-resolved half-reaction, including catalyst site labels, molecular oxygen, water, and electrons.



答案

$$\mathcal{C}[^{16}\mathrm{O},{}^{16}\mathrm{O}] + 4\,{}^{18}\mathrm{OH}^- \rightarrow \mathcal{C}[^{18}\mathrm{O},{}^{18}\mathrm{O}] + {}^{16}\mathrm{O}_2 + 2\mathrm{H}_2{}^{18}\mathrm{O} + 4e^-$$

解析

The dual-site oxygen evolution cycle couples both blue lattice oxygen atoms before molecular oxygen is released. The gas therefore receives two initial lattice isotopes and no electrolyte isotope.

$$n_{^{16}\mathrm{O},\mathrm{gas}} = 2, \qquad n_{^{18}\mathrm{O},\mathrm{gas}} = 0.$$

The molecular oxygen isotopologue is consequently fixed.

$$\mathrm{O}_{2,\mathrm{gas}} = {}^{16}\mathrm{O}_2.$$

Release of the paired lattice oxygen unit empties both tracked sites.

$$n_{\mathrm{vac}} = 2.$$

Hydroxide-only refill changes both tracked catalyst sites from the initial isotope to the electrolyte isotope.

$$\mathcal{C}[^{16}\mathrm{O},{}^{16}\mathrm{O}] \longrightarrow \mathcal{C}[^{18}\mathrm{O},{}^{18}\mathrm{O}].$$

A four-electron alkaline turnover consumes four hydroxides. Two of their oxygen atoms refill the lattice, so isotope conservation assigns the other two electrolyte oxygen atoms to water.

$$4 = n_{^{18}\mathrm{O},\mathrm{lat}} + n_{^{18}\mathrm{O},\mathrm{water}} = 2 + n_{^{18}\mathrm{O},\mathrm{water}}, \qquad n_{^{18}\mathrm{O},\mathrm{water}} = 2.$$

The four hydroxide hydrogens then fix the water coefficient.

$$4 = 2n_{\mathrm{H_2O}}, \qquad n_{\mathrm{H_2O}} = 2.$$

Charge conservation fixes the electron coefficient on the product side.

$$-4 = -n_e, \qquad n_e = 4.$$

Combining gas formation, two-site isotope exchange, water formation, and electron balance gives the isotope-resolved half-reaction.

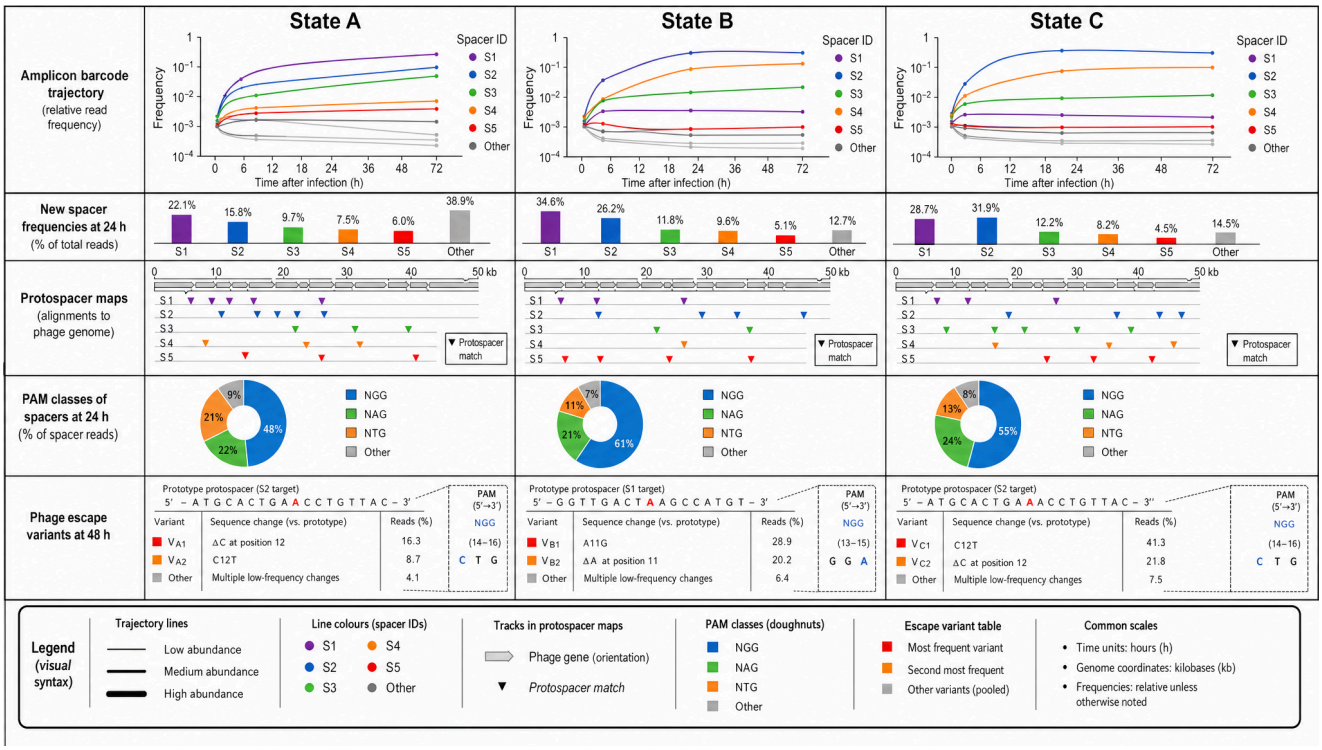
$$\mathcal{C}[^{16}\mathrm{O},{}^{16}\mathrm{O}] + 4\,{}^{18}\mathrm{OH}^- \rightarrow \mathcal{C}[^{18}\mathrm{O},{}^{18}\mathrm{O}] + {}^{16}\mathrm{O}_2 + 2\mathrm{H}_2{}^{18}\mathrm{O} + 4e^-.$$

类目

Biology / CRISPR adaptation and phage escape

题目

Read the printed assay times for spacer/PAM sampling and escape-variant sampling, and use the oriented genome track to decide which PAM-side factor applies. The phage-gene arrows define downstream along the genome track; a PAM callout on the downstream flank uses $q = 1.25$, whereas a PAM callout on the upstream flank uses $q = 0.80$. The State A, State B, State C order is the column order. For a serial CRISPR adaptation model, use $x = e(1 - c) = (0.3604, 0.1536, 0.1421)$, $r = (0.61, 0.38, 0.29)$, $c = (0.32, 0.36, 0.71)$, $B = 4.2$, $\delta = 0.18\,\mathrm{h}^{-1}$, and $\kappa = 0.14\,\mathrm{h}^{-1}$. Acquisition memory is $M_i(\theta) = 1 - \exp[-\theta q x_i t_s]$. If spacer/PAM sampling precedes escape-variant sampling, the fitted PAM-constrained escape burden is $H_i(\theta) = Z_i[1 - M_i(\theta)]/[1 + qM_i(\theta)]$, where $Z_i = Br_i(1 - \exp[-\delta\Delta t])\exp[-\kappa c_i \Delta t]$ and Δt is the elapsed time between those two readouts; if the escape-variant readout did not follow spacer/PAM sampling, this post-acquisition interference denominator would not be applied. A matched bulk assay gives $H_A + H_B + H_C = 0.06078$. Which State has the largest fitted H_i, and what is its fold excess over the smallest fitted H_i, rounded to two decimals?



答案

(State B, 5.95-fold)

解析

The spacer-frequency and PAM-class rows are timed at 24 h, while the escape-variant row is timed at 48 h. That makes spacer acquisition the earlier process and places the interference competition term after acquisition in the serial burden model.

$$\Delta t = 48\,\mathrm{h} - 24\,\mathrm{h} = 24\,\mathrm{h}.$$

The genome-track arrows run to the right and the PAM callout lies to the right of the prototype protospacer sequence area, so the stated flank convention uses the downstream PAM factor rather than the upstream one.

$$q = 1.25.$$

Separate the phage expansion and clearance part from the acquisition-filtering part by defining the escape prefactor Z_i. Using the elapsed 24 h interval gives the three prefactors in State A, State B, State C order.

$$Z_i = B r_i \left(1 - e^{-\delta \Delta t}\right) e^{-\kappa c_i \Delta t}.$$

This term is not just a scaling constant: it combines phage replication with clearance over the interval after spacer/PAM sampling and before escape-variant readout.

$$Z = (0.8631,\ 0.4698,\ 0.1106).$$

Let $s_i(\theta) = 1 - M_i(\theta) = \exp[-\theta q x_i t_s]$. The fitted burden therefore has both an acquisition survival numerator and a PAM-guided interference denominator.

$$H_i(\theta) = \frac{Z_i s_i(\theta)}{1 + q[1 - s_i(\theta)]}.$$

Substituting $t_s = 24\,\mathrm{h}$, $q = 1.25$, and $x = (0.3604, 0.1536, 0.1421)$ gives the coupled bulk-fit equation.

$$\frac{0.8631 e^{-10.812\theta}}{2.25 - 1.25 e^{-10.812\theta}} + \frac{0.4698 e^{-4.608\theta}}{2.25 - 1.25 e^{-4.608\theta}} + \frac{0.1106 e^{-4.263\theta}}{2.25 - 1.25 e^{-4.263\theta}} = 0.06078.$$

The left side is a decreasing sum of positive terms, so the fitted acquisition coefficient is unique for the chosen timing and PAM flank.

$$\hat{\theta} = 0.370\,\mathrm{h}^{-1}.$$

At this coefficient, the state-specific fitted burdens are obtained by substituting back into the same post-acquisition interference model.

$$H = (0.00710,\ 0.04222,\ 0.01147).$$

The largest burden is therefore in State B, and the smallest is in State A.

$$\frac{H_B}{H_A} = \frac{0.04222}{0.00710} = 5.95.$$

If the escape readout were not later than the spacer/PAM readouts, the post-acquisition denominator and the expansion interval would not be the same, so the bulk fit would not produce these H_i values. If the PAM callout were on the upstream flank, q would change in both the acquisition exponent and the interference denominator, changing the fitted coefficient and the fold excess. With the observed timing, genome orientation, and PAM-side relation, the closed result is $(\mathrm{State\ B}, 5.95\text{-fold})$.

类目

Biology / Axonal transport and molecular motors

题目

The three State columns show matched axonal cargo transport assays. Use the kymograph key and the process-state legend to instantiate the following hidden Markov delivery model. The initial prior masses assigned to the process-diagram node labels $A, B, R, 0$, in that order, are

$$\pi = (0.31, 0.29, 0.25, 0.15).$$

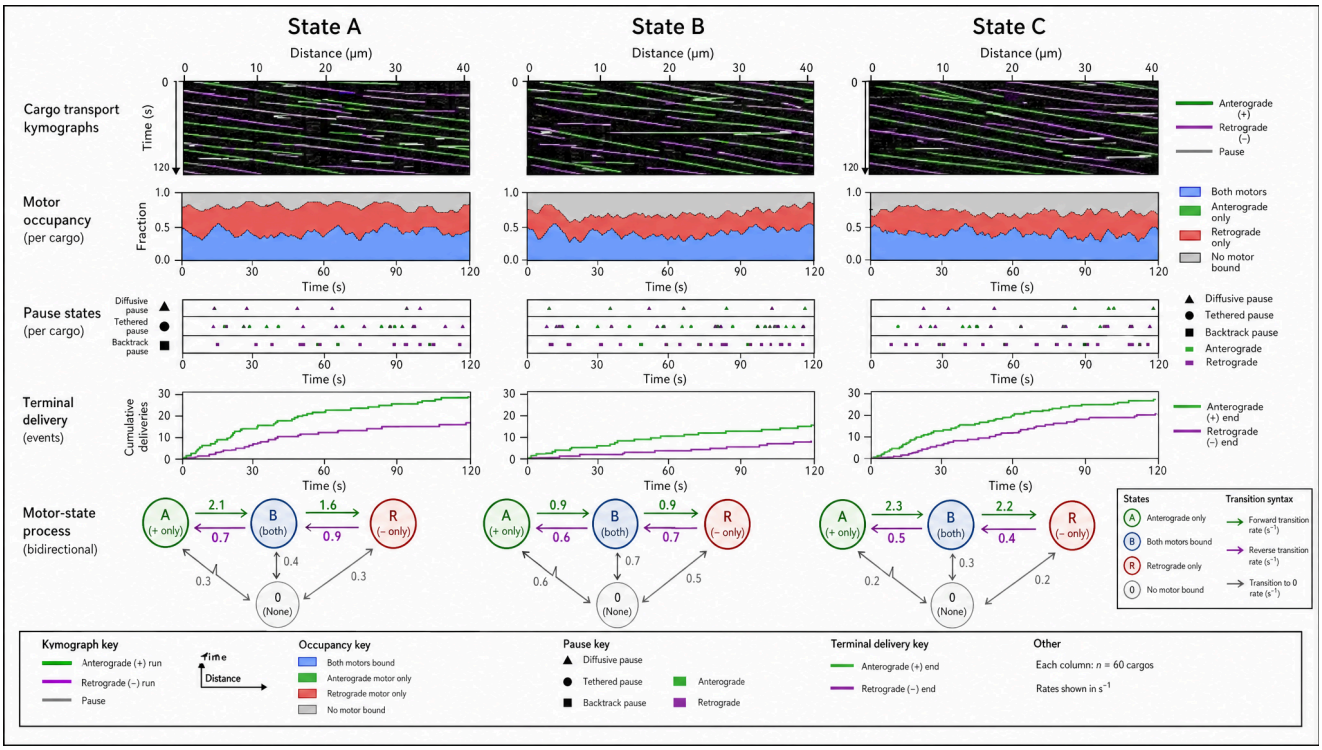
Normalize over motor-bound cargos after removing the node identified by the legend as no motor bound. For the motor-bound vector ordered as $+$-only, both-bound, $-$-only, the run-pause switching operator for a reversible bidirectional run-pause phenotype is

$$S = \begin{bmatrix} 0.62 & 0.08 & 0.03 \\ 0.28 & 0.72 & 0.27 \\ 0.10 & 0.20 & 0.70 \end{bmatrix},$$

where columns are the pre-switch motor state and rows are the post-switch motor state. Apply S before terminal-delivery readout only when every State column's kymograph contains both directed run categories and pause segments; otherwise omit S. The both-bound tug-of-war readout is set by

$$\rho = \frac{1}{1 + \exp[-5.8(0.73 - 0.69)]}, \qquad d_T = 0.18 + 0.64\rho.$$

The single $+$-only and single $-$-only terminal readouts are $d_+ = 0.82$ and $d_- = 0.18$. The interval completion factor is $m = 1 - \exp(-40/18)$. Using the cargo number printed in the legend, what is the expected number of anterograde terminal deliveries in one State column, rounded to one decimal cargo?



答案

26.8 cargos

解析

Green and purple diagonal kymograph streaks occur together with gray pause segments in each of the three State columns. The keys identify these as anterograde + runs, retrograde − runs, and pauses, so the reversible run-pause operator is applied before terminal readout rather than omitted.

$$\pi_{\mathrm{bound}} = 1 - \pi_0 = 1 - 0.15 = 0.85.$$

The process-state legend identifies the unbound node as 0, so the hidden-state prior is first conditioned on motor-bound cargos. The printed process labels identify A as +-only, B as both-bound, and R as −-only.

$$\tilde{\pi} = \left(\frac{0.31}{0.85}, \frac{0.29}{0.85}, \frac{0.25}{0.85}\right) = (0.3647, 0.3412, 0.2941).$$

Because the matched kymographs show both directions plus pauses, the bound prior is propagated through the switching matrix before the terminal-delivery probabilities are read out.

$$x = S\tilde{\pi} = \begin{bmatrix} 0.62 & 0.08 & 0.03 \\ 0.28 & 0.72 & 0.27 \\ 0.10 & 0.20 & 0.70 \end{bmatrix} \begin{bmatrix} 0.3647 \\ 0.3412 \\ 0.2941 \end{bmatrix} = (0.2622, 0.4272, 0.3106).$$

The both-motor node is treated as a tug-of-war state, so its anterograde terminal readout is nonlinear rather than equal to either single-motor readout.

$$\rho = \frac{1}{1 + \exp[-5.8(0.04)]} = 0.5577, \qquad d_T = 0.18 + 0.64(0.5577) = 0.5369.$$

The post-switch terminal probability combines the $+$-only, both-bound, and $-$-only readouts in the post-switch mixture.

$$p_{+\text{end}|\text{switch}} = 0.82(0.2622) + 0.5369(0.4272) + 0.18(0.3106) = 0.5003.$$

Only a completed 40 ms interval contributes to terminal delivery, giving the interval-scaled probability per cargo.

$$p_{\text{deliver}} = (1 - \exp[-40/18])(0.5003) = 0.4461.$$

The legend states that each State column contains 60 cargos, so the expected anterograde terminal count is

$$E[N_+] = 60(0.4461) = 26.77 \approx 26.8 \text{ cargos.}$$

If one State column lacked either directed run category or lacked pauses, S would be omitted and the expected count would increase to about 28.6 cargos. If the both-bound or unbound process node were changed, the normalization and nonlinear readout would be applied to different hidden states, giving a different count. The label and kymograph patterns therefore fix a single calculation path.